7.1 - Networks and Trees

Problem Solving, Math 1000

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Question

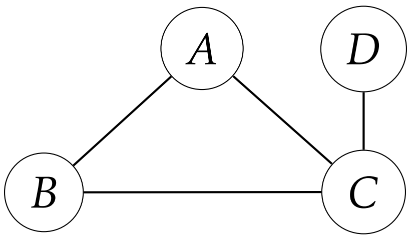
When people talk about networks – things like wireless networks, social networks, political networks – what do they mean? Is this a mathematical concept at its heart? If so, can we use mathematical techniques to better describe, quantify, and yield insight into these real-world phenomena?

Definition

A network is just another name for a connected graph. A simple network is a simple, connected graph. This isn't a new concept in any meaningful sense, just a new way to refer to a familiar concept to make things easier.

Example

The following graph is an example of a simple network:

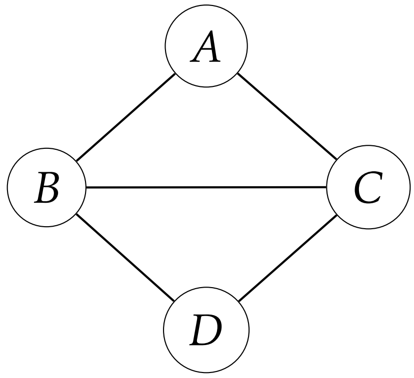


Definition

The degree of separation of two vertices in a network is the length of the shortest path joining those vertices.

Example

The degree of separation of A and D in the graph below is 2. The degree of separation of B and C is 1.

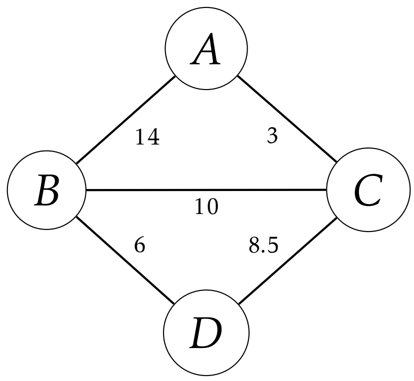


Definition

A weighted network is a network where each edge is a assigned a numerical value called a weight. Generally, this is depicted in a drawing of a graph that is a network by writing a number next to each edge.

Example

The following weighted graph is an example of a weighted network



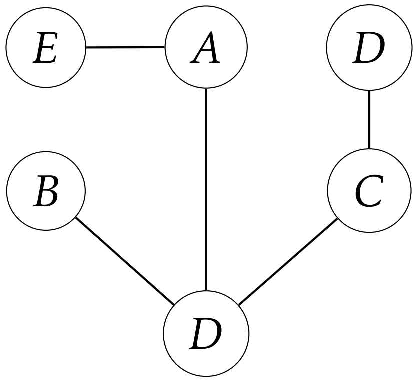
Definition

A network is a tree if it contains no circuits. Trees have a variety of interesting properties:

1. A tree has one and only one path between any two vertices. That is to say, each pair of vertices defines a unique path.
2. In a tree, every edge in the graph is a bridge.
3. A tree with n vertices has $n-1$ edges. Conversely, any network with n vertices and $n-1$ edges must be a tree graph.

Example

The following network is a tree.



Definition

In a network with n vertices and m edges, the redundancy, denoted R, of the network is defined to be $R = m-(n-1)$. This follows from the fact that $n-1$ edges (i.e. the number of edges in any tree on n vertices) is the minimal number of edges necessary to connect the graph (that is, any graph with n vertices and fewer than $n-1$ edges cannot be a network). This gives us the characterization that a network with zero redundancy (i.e. $R = 0$) is a tree.

Conclusions

We may rephrase the above by the following key points:

1. A tree is a minimally connected network. That is, no edge may be removed from a tree without disconnecting the graph (every edge is a bridge), and the network has no redundancy. So, if you want to minimize the number of edges in a network, you build a tree.

2. A network that is not a tree has some redundant edges (i.e. $R \neq 0$). These redundant edges form circuits, which allow for multiple ways to ”get around” the graph, so to speak. So, if you want a network with many alternate routes, you build one with a large degree of redundancy.